

AMIT KUMAR

+91 7358325664

Remote

amitk.softtech@gmail.com

in /amitkumarcse09

Professional summary

Principal Software Engineer with 10+ years of experience in building and modernizing enterprise software platforms. Expertise in fullstack development thru Java, Spring Boot, React, distributed systems and ETL technologies, with a proven record of leading architecture modernization, enterprise product releases and cross-functional engineering teams. Delivered multiple Pentaho platform suite releases, led migration, modernization and improved Enterprise Edition adoption through customer-driven product enhancements. Passionate about building scalable, high-performance software while mentoring engineers and driving technical excellence.

Career highlights

- **10+ years** of enterprise engineering experience across **Data Processing Platform, FinTech & Defence** domains.
- Functioning as **Principal Software Engineer** leading platform modernization for **Pentaho**.
- Technical lead in a 26-member engineering organization with ownership of a **12-member core engineering team**.
- **Release Manager** for Pentaho Enterprise 10.0 and 11.1, driving major releases and service-pack delivery.
- Led Pentaho Java modernization from Java 8 to Java 17, strengthening enterprise and hyperscaler compatibility.
- Modernized Pentaho's plugin architecture .
- Increased **Enterprise Edition trial-to-paid conversion by 22%** through customer-centric product enhancements.
- Delivered **200+ React components** and **1000+ backend services/plugins** for enterprise-scale ETL solutions.
- Strong expertise in software **architecture, technical leadership, customer engagement, and AI-assisted software engineering**.
- Early adopter of **AI-powered software engineering** practices, utilizing **GitHub Copilot**, Claude, LLMs, prompt engineering, custom AI agents, and workflow automation thru LanGraph to improve software delivery, engineering quality, and team productivity.

Technical skills

Programming Languages

- Java (8/11/17)
- TypeScript
- JavaScript
- SQL
- Shell Scripting

Backend

- Spring Boot
- Spring MVC
- REST APIs
- Microservices
- Concurrency
- Executor Framework

Distributed Systems

- Apache Kafka
- ZooKeeper
- Redis
- Distributed Caching
- Apache Solr
- Apache Nutch

Build tools & DevOps

- Linux
- Git
- TeamCity
- UDeploy
- SonarQube
- Jenkins

Software Architecture

- Microservices
- Plugin Architecture
- Design Pattern
- Debugging
- Profiling

Engineering Leadership

- Architecture Reviews
- Code Reviews
- Sprint Planning
- Release Management
- Jira

Professional experience

Software Development Eng. Specialist Professional (p4) / Principal SDE Hitachi Vantara

April 2022 - Present

Pentaho Data Integration (PDI) is Hitachi Vantara's enterprise-grade data integration and analytics platform that enables organizations to design, orchestrate, and execute scalable ETL/ELT pipelines across on-premises, cloud, and hybrid environments. The platform provides seamless integration with relational databases, cloud storage, streaming technologies, big data ecosystems, and enterprise applications, and is distributed across desktop, server, and cloud-ready deployments for global enterprise customers.

My role is to drive platform architecture, modernization, and end-to-end product development across the Pentaho ecosystem. My responsibilities include designing scalable solutions, leading Java platform modernization, evolving the plugin framework, developing enterprise features across backend and React-based frontend applications, managing product releases, collaborating directly with enterprise customers, mentoring engineers, and leading cross-functional teams to deliver high-quality enterprise software.

Key Responsibilities & Technical Contributions:

Function as a **Principal Software Engineer** (official designation: **Specialist Professional – Software Development Engineer**) responsible for driving architecture, platform modernization, and enterprise product development for Pentaho Data Integration (PDI).

Lead end-to-end software development lifecycle, including solution design, architecture, development, testing, release management, deployment, and production support.

Provide technical leadership to a **26-member engineering organization**, leading a **12-member core engineering team** across multiple product initiatives.

Drive architecture discussions and technical decision-making for enterprise platform modernization and new feature development.

Collaborate with Product Management, QA, Customer Success, and enterprise customers to translate business requirements into scalable product capabilities.

Lead sprint planning, backlog grooming, Jira epic and story creation, effort estimation, release planning, and engineering execution.

Conduct architecture reviews, design reviews, pull request reviews, and enforce engineering best practices across the development team.

Mentor engineers through technical guidance, knowledge-sharing sessions, debugging support, and code quality improvements.

Own production issue analysis, root cause investigation, customer escalations, and timely resolution of critical defects.

Ensure engineering excellence through unit testing, static code analysis, SonarQube quality gates, and continuous integration practices.

Platform Modernization

Spearheaded the migration of the Pentaho platform from **Java 8 → Java 11 → Java 17**, modernizing the codebase while ensuring backward compatibility and platform stability.

Led compatibility validation and certification of the platform across supported hyperscaler environments as part of the Java modernization initiative.

Modernized legacy dependencies and resolved compatibility issues introduced by newer Java runtime versions.

Refactored legacy components to align with modern Java development standards and improve long-term maintainability.

Platform Architecture

Led the modernization of Pentaho's plugin framework by replacing the legacy **OSGi-based architecture** with a lightweight dynamic plugin architecture.

Designed and implemented a custom **Java ClassLoader** enabling runtime plugin discovery and dynamic loading of plugins.

Eliminated OSGi runtime dependency, reducing platform complexity and simplifying deployment and maintenance.

Collaborated with architects and engineering teams to ensure seamless migration of existing plugins to the new architecture.

Product Engineering

Designed and developed enterprise-grade backend services and product features supporting the Pentaho ecosystem.
Developed **200+ React components** and **1000+ backend services/plugins** for enterprise ETL and data integration workflows.
Designed and implemented enterprise plugins including **Kafka, Database Lookup, Combination Lookup, Dimension Lookup, Python Executor, Excel Processing, and Cloud Storage Connectors**.
Developed storage connectors supporting virtual file systems and cloud storage providers including **Amazon S3**.
Designed scalable backend APIs and business logic supporting enterprise-grade ETL processing.
Built reusable frontend components using React and TypeScript to modernize legacy desktop functionality.

Release Engineering
Contributed to the successful delivery of **Pentaho Enterprise 9.3, 9.4, 10.0, 10.1, 11.0, and 11.1**.
Served as **Release Manager** for **Pentaho Enterprise 10.0** and **11.1**, owning end-to-end release coordination and delivery.
Managed release branch strategy, branch cut, code integration, stabilization, release readiness, and production sign-off.
Governed release quality through code reviews, SonarQube analysis, automated testing, and engineering quality gates.

Customer Success & Business Impact
Worked directly with enterprise customers to understand business challenges and translate requirements into product capabilities.
Delivered customer-driven product enhancements that increased **Enterprise Edition trial-to-paid conversion by 32%**.
Partnered with Customer Success and Product Management teams to accelerate enterprise customer onboarding and product adoption.

Installer Modernization
Architected and implemented seamless **upgrade and downgrade** capabilities across Pentaho installer distributions (**.exe, .tar.gz, .zip, and .jar**).
Eliminated the need for complete uninstall and reinstallation during version upgrades, significantly improving the customer deployment experience.

AI-Assisted Engineering
Leveraged **GitHub Copilot, LLM-based assistants, prompt engineering, and AI-driven workflows** to accelerate software development, code reviews, refactoring, documentation, and engineering productivity.

Senior Associate – Technology
Synechron
May 2021 – Mar 2022

On boarded RBI regulator, CCIL on CITI regulatory platform

Conceptualized business regulatory requirements by collaborating with government officials from Reserve Bank of India and CITI technology upstream and downstream teams. Finalized and outlined the goals and expectations of CITI business group.
Delivered BRD and FSD. Recreated API contract document.
Created application architecture and skeleton on Spring Boot and integrated CI/CD pipeline with TeamCity and UDeploy.
Proposed fall-back plan and application failover mechanism.
Created required DB tables and shell scripts.
Coded and unit tested services written in java 8.
Drove project to production on proposed timeline.

Tech stack used: Spring Boot 2.5.3, Spring Core & MVC, Java 8, JDBC template, Apache POI, REST api & curl, Algorithms, JSON, Teamcity, Udeploy, Network proxy system

Modules written:

- Directory poller / File watcher with custom configuration.
- CSV to MS XLS file conversion thru apache POI library. Adhering to format provided by regulator.
- Preparing API payload in JSON format. Implementing API contract document.
- API calls thru REST. Implementing API call sequence.
- Casting response received from regulator from JSON to object format.
- Response writing to MS XLS file in separate sheet of same XLS file.
- Automated e-mail to concerned team thru spring mail.
- Persisting transactions in Oracle table for audit purpose using JDBC template.

- Prepare and send end of day report of day in out requests. Implemented Executor Service framework to read multiple MS XLS files and write to single XLS file.
- Manage periodic API password updates thru AES 256 cipher and decipher within application properties. Calling job thru AutoSYS tool. Generating new password as per regulator policy using coded algorithm.
- Handle exception and failure scenarios.
- **Most challenging task:** Third party API integration on publicly exposed network.

Extra miles deliveries:

- Created SQL triggers on Oracle table to start data processing for APAC region regulators.
- Coded module for FPML/XML value updation based on tag hierarchy.
- Actively participated and contributed to CRB & production releases.
- Mentored 3 associates for technical guidance over Java & shell scripting.
- Co-ordinated monthly BAU production release as primary resource.

Executive engineer

Bharat Electronics Ltd.

/DRDO, Ministry Of Defence

Oct 2018 – May 2021

Open Source Intelligence – OSINT

Developed data center lab to gather data available over internet on various platforms like webpages, twitter, youtube, facebook & other govt. platforms.

Create scalable data centre & process, store & retrieve data on demand.

Collaborate with various Defence Intelligence Agencies to understand and cater their demand thru intelligent search engine & dashboards.

Tech stack used: Spring Boot 2.5.3, Spring Core & MVC, Java 8, JDBC template, Apache POI, REST api & curl, Algorithms, JSON, Teamcity, Udeploy, Network proxy system

Modules written:

- Spring boot applications in micro service architecture. Each module is thread safe and have low latency.
- Implemented failover mechanisms at module and code level. Also implemented Swagger, security, logging & actuator.
- Spring data implementation to store user and request data into MongoDB.
- Created custom schema for SOLR document. Schema depending upon platform data and requirement.
- Build data fetch pipeline from SOLR upon incoming query using solr4j.
- Customising Apache Nutch crawler to fetch Lat-Long coordinates & location name from 3rd party API. Also included new features of removing stopper words.
- Scrape meta search engines like Baidu, Yandex, Google, DuckduckGo, Yahoo, etc. using jsoup lib to prepare seed list to start internet crawling. User should be capable to crawl thru deep web and dark web .onion websites also.
- Established micro service communication thru API contract. Later converted into Kafka MQ.
- Led a team of 5 associates and delivered product on proposed timeline.

Distributed system development skills

- Created distributed system on [Debian 9 OS](#) thru CLI. Deb package used :[Oracle virtual box](#) (vboxmanage) & [Gpartition](#).
- Configured newly created systems [to intranet network](#) by assigning IP, DNS, Proxy creds. & system variables.
- Enable [SSH passwordless login](#) within newly created [cluster](#).
- Installed required deb. Packages to each system & classifying them into commodity H/W & Master H/W.
- Installed distributed setup of [Apache Zookeeper\(3.14\)](#), [Hadoop\(2.5\)](#), [Hbase\(1.2\)](#), [Apache Solr\(8.0\)](#), [MongoDB](#), [TOR & Provoxy](#). And [changed configuration as root user](#) as per requirements.
- Setup [NGINX](#) and configured load balancer configs. And deploy [JAR](#) to make system ready for service.
- Create shell script to [install this setup without manual intervention](#) on a freshly created VMs.

Software Engineer

Fidelity Investments

Aug 2015 – Oct-2018

Adhoc Information Management System

Delivering verity of compliance reports to business clients on periodic basis and updating records in batch.

Convert business language request into technical specifications.

Identify data source for the required report and plan strategy to consolidate data to XLS or CSV format.

Write SQL code (mostly), Mainframe COBOL / Easytrieve code to pull data from cluster of DB2 tables, VSAMs, flat files & historical tapes & use intermediate common format to store. Mostly in MS access DB.

- Do data massaging, data cleansing and eliminate unnecessary information from data and cleans a dataset to make it useable. It involved processing data to change data formats, remove unwanted characters, duplicates, whitespaces, and more
- Frequency of request to be 5/day (avg.). Communicate effectively with client for clarifications or concerns
- Receive Batch update requests from business clients to update numerous fields in accounts.
- Convert requests into program consumable formats and perform change against prod similar environment thru COBOL and java batch jobs.
- Validate results numerically and logically and stage file for production to nightly batch jobs.
- Code and automate most frequent requests type as java job to deliver requests without manual intervention.

Educational qualification

MBA – online 2022-2024

specialization: Project Management

D.Y Patil Vidhyapeeth, Pimpri, Pune MH

Bachelor of Technology: Computer Science & Engineering 2010-2014

Arya College of Engineering and Research Centre, Jaipur, Rajasthan,

Rajasthan Technical University (RTU, Kota)

12th: Kendriya Vidyalaya, Vayusena Nagar, Nagpur, Maharashtra 2010

10th: Kendriya Vidyalaya, Vayusena Nagar, Nagpur, Maharashtra 2008

Other details:

D.O.B: 09-Mar-1993

Gender: Male

Language proficiency: English, Hindi

Above presented information is best of my knowledge.

(Amit Kumar)